Audit trail – Thematic synthesis on uncertainties in healthcare

Initial coding in N-vivo – 107 codes identified							
Name	**Files**	**References**	**Memo Link**	**Created By**	**Created On**	**Modified By**	**Modified On**
1. Unpredictability leading to uncertainties	4	4		E	8/4/2021 5:46 PM	E	8/6/2021 5:46 PM
2. Statistical uncertainty	1	1		E	7/4/2021 6:14 PM	E	7/6/2021 6:18 PM
3. Future risk leading to uncertainties	3	3		E	7/4/2021 6:23 PM	E	8/6/2021 7:40 PM
4. Expensive investigations	4	7		E	4/4/2021 8:39 AM	E	8/6/2021 7:38 PM
5. Lack of robust clinical protocols	4	6		E	3/4/2021 11:07 AM	E	8/6/2021 7:38 PM
6. Resemblance to other diseases	2	3		E	4/4/2021 8:43 AM	E	8/6/2021 7:37 PM
7. Role of co-factors	1	1		E	7/4/2021 6:18 PM	E	8/6/2021 7:33 PM
8. Seriousness of condition	2	3		E	3/4/2021 10:03 AM	E	5/6/2021 3:30 PM
9. Negotiation skills	1	1		E	23/3/2021 8:17 AM	E	8/6/2021 7:35 PM
10. Uncertainties due to cues picked during conversation	2	4		E	2/4/2021 11:55 AM	E	8/6/2021 7:36 PM
11. Uncertainties due to language used during communication	2	2		E	23/4/2021 8:18 AM	E	8/6/2021 7:32 PM
12. Uncertainty due to lack of patient-centred communication	3	6		E	8/4/2021 5:51 PM	E	8/6/2021 5:51 PM
13. Clinicians' ignorance or lack of skill	2	2		E	8/4/2021 5:51 PM	E	8/6/2021 5:51 PM
14. Patient's level of education	1	2		E	5/4/2021 3:26 PM	E	8/6/2021 7:45 PM
15. Ambiguity leading to uncertainties	3	4		E	4/4/2021 11:01 AM	E	10/6/2021 7:43 PM
16. Complexity leading to uncertainties	6	6		E	4/4/2021 9:51 AM	E	10/6/2021 7:42 PM
17. Lack of credibility leading to uncertainties	1	3		E	5/4/2021 3:30 PM	E	10/6/2021 4:42 PM
18. Growth of medical technology related knowledge leading to uncertainties	1	2		E	3/5/2021 2:27 AM	E	10/6/2021 7:48 PM

Audit trail – Thematic synthesis on uncertainties in healthcare

Initial coding in N-vivo – 107 codes identified							
Name	**Files**	**References**	**Memo Link**	**Created By**	**Created On**	**Modified By**	**Modified On**
19. Less knowledge or information leading to uncertainties	4	6		E	8/4/2021 6:38 PM	E	10/6/2021 10:06 AM
20. More knowledge or information leading to uncertainties	4	4		E	4/4/2021 3:05 PM	E	11/6/2021 7:27 PM
21. Prioritising patients due to lack of resources	2	2		E	23/4/2021 11:10 AM	E	10/6/2021 3:22 PM
22. Time pressure dictating patient treatment	1	1		E	8/4/2021 5:52 PM	E	18/6/2021 5:52 PM
23. To inform or not to inform patients about lack of evidence	1	1		E	4/4/2021 3:25 PM	E	10/6/2021 7:57 PM
24. Clinicians making decisions on behalf of the patient or family members	1	2		E	5/4/2021 9:01 AM	E	15/6/2021 7:58 PM
25. Parents making decisions on behalf of their child	2	3		E	13/4/2021 12:34 PM	E	18/6/2021 7:58 PM
26. Suddenness of the situation	1	1		E	13/4/2021 3:29 PM	E	18/6/2021 7:59 PM
27. Uncertainty due to lack of conclusive results from emergency research	1	1		E	13/4/2021 5:53 PM	E	18/6/2021 5:53 PM
28. Uncertainties created by media	1	3		E	13/4/2021 12:35 PM	E	19/6/2021 8:00 PM
29. Uncertainties due to social arrangement of power	1	1		E	13/4/2021 7:18 PM	E	18/6/2021 8:00 PM
30. Uncertainties due to subjective decision making	1	1		E	16/4/2021 10:33 AM	E	18/6/2021 8:00 PM
31. Inability to establish a cause-and-effect relationship	1	1		E	13/4/2021 1:52 PM	E	15/6/2021 3:27 PM
32. Uncertainties due to lack of pattern in disease presentation	1	2		E	13/4/2021 1:09 PM	E	20/6/2021 8:01 PM

Audit trail – Thematic synthesis on uncertainties in healthcare

Initial coding in N-vivo – 107 codes identified							
Name	**Files**	**References**	**Memo Link**	**Created By**	**Created On**	**Modified By**	**Modified On**
33. Timing of the week or day	1	1		E	13/4/2021 3:26 PM	E	22/6/2021 8:02 PM
34. Errors in reporting	1	1		E	20/4/2021 8:22 AM	E	16/6/2021 6:13 PM
35. Over testing in response to or leading to uncertainties	1	2		E	20/4/2021 5:51 PM	E	23/6/2021 5:06 PM
36. Lack of check mechanism	2	2		E	22/4/2021 8:44 AM	E	16/6/2021 5:08 PM
37. Lack of facilities	3	5		E	20/4/2021 8:43 AM	E	25/6/2021 5:07 PM
38. Level of experience	1	3		E	20/4/2021 5:47 PM	E	25/6/2021 5:47 PM
39. Clinicians' personal fears	2	1		E	20/4/2021 5:49 PM	E	25/6/2021 5:50 PM
40. Previous experiences (clinician)	1	2		E	20/4/2021 3:41 PM	E	28/6/2021 3:25 PM
41. Proxy uncertainties (clinician)	1	1		E	22/4/2021 9:11 AM	E	8/7/2021 7:56 PM
42. Risk taking behaviour (Clinician)	3	3		E	20/4/2021 5:48 PM	E	25/6/2021 7:55 PM
43. Uncertainty intolerance (Clinician)	2	3		E	22/4/2021 11:06 AM	E	28/6/2021 7:54 PM
44. Unclear patient characteristics	1	1		E	22/4/2021 9:00 AM	E	28/6/2021 7:54 PM
45. Influence of friends and family	3	5		E	22/4/2021 12:37 PM	E	25/6/2021 7:54 PM
46. Patient's level of education	1	1		E	22/4/2021 8:15 AM	E	25/6/2021 7:54 PM
47. Patients' personal beliefs	5	8		E	22/4/2021 5:33 PM	E	25/6/2021 5:33 PM
48. Patients' personal fears	5	4		E	26/4/2021 5:36 PM	E	25/6/2021 5:36 PM
49. Risk taking behaviour	2	3		E	4/4/2021 8:56 AM	E	8/7/2021 7:21 PM
50. Trust issues	1	1		E	4/4/2021 8:56 AM	E	8/7/2021 7:21 PM
51. Uncertainty negatively perceived	3	3		E	8/4/2021 5:37 PM	E	25/6/2021 6:47 PM
52. Uncertainty positively perceived (coping with)	8	8		E	23/4/2021 12:24 PM	E	28/6/2021 7:22 PM
53. Proxy uncertainties (Carers)	6	8		E	22/4/2021 9:56 AM	E	23/6/2021 5:45 PM
54. Associated fears of parents	10	15		E	4/4/2021 9:22 AM	E	28/6/2021 7:23 PM
55. Personal interpretations (parents)	1	1		E	22/4/2021 5:35 PM	E	28/6/2021 5:36 PM

Audit trail – Thematic synthesis on uncertainties in healthcare

Initial coding in N-vivo – 107 codes identified							
Name	**Files**	**References**	**Memo Link**	**Created By**	**Created On**	**Modified By**	**Modified On**
56. Lack of quantifiability or measurability	3	4		E	2/5/2021 6:54 PM	E	28/6/2021 11:46 AM
57. Model inadequacy	3	3		E	2/5/2021 6:56 PM	E	28/6/2021 7:07 PM
58. Procedural uncertainties	2	2		E	2/5/2021 5:23 PM	E	29/6/2021 5:23 PM
59. Guidelines based on evidence which cannot be generalised	1	1		E	2/5/2021 5:07 PM	E	29/6/2021 3:14 PM
60. Inability to apply guidelines to all situations	1	1		E	4/4/2021 10:59 AM	E	29/6/2021 9:24 PM
61. Uncertainties due to socio-economic inequalities and inequities	1	1		E	2/5/2021 9:22 PM	E	30/6/2021 9:44 AM
62. Uncertainties in implementing preventive measures	3	3		E	2/5/2021 9:20 PM	E	29/6/2021 9:25 PM
63. Uncertainty due to lack of standardised definitions	2	2		E	5/5/2021 9:21 PM	E	1/7/2021 9:45 AM
64. Uncertainties due to challenges in data collection	1	1		E	9/5/2021 9:24 PM	E	1/7/2021 10:04 AM
65. Uncertainties due to previous faulty reporting	2	4		E	4/5/2021 8:30 AM	E	1/7/2021 9:46 AM
66. Uncertainty due to unclear risk distribution	1	1		E	23/5/2021 12:26 PM	E	29/6/2021 5:48 PM
67. Interaction of clinicians with other health care team leading to uncertainties	1	1		E	8/5/2021 5:09 PM	E	29/6/2021 3:12 PM
68. Interaction or conflict between patients' friends, relatives with clinician team leading to uncertainties	2	2		E	23/5/2021 9:11 AM	E	9/7/2021 5:51 PM
69. Physician- patient interaction leading to uncertainties	1	3		E	4/5/2021 9:38 AM	E	9/7/2021 5:51 PM
70. Improper team management leading to uncertainties	3	3		E	4/5/2021 8:22 PM	E	9/7/2021 5:52 PM

Audit trail – Thematic synthesis on uncertainties in healthcare

Initial coding in N-vivo – 107 codes identified							
Name	**Files**	**References**	**Memo Link**	**Created By**	**Created On**	**Modified By**	**Modified On**
71. Trust issues in the team leading to uncertainties	1	3		E	23/5/2021 9:18 AM	E	9/7/2021 5:53 PM
72. Uncertainties due to avoiding referral	3	4		E	4/5/2021 10:56 AM	E	9/7/2021 5:54 PM
73. Uncertainties due to involvement of multiple people	6	8		E	26/5/2021 5:55 PM	E	9/7/2021 5:54 PM
74. Uncertainties due to referral	3	5		E	23/5/2021 9:14 AM	E	9/7/2021 5:55 PM
75. Uncertainties due to seeking help from others	1	1		E	23/5/2021 9:16 AM	E	9/7/2021 5:56 PM
76. Fear of stigma leading to uncertainties	2	2		E	24/5/2021 5:09 PM	E	9/7/2021 3:13 PM
77. Helplessness or feeling unsupported leading to uncertainties	2	2		E	26/5/2021 6:09 PM	E	9/7/2021 5:57 PM
78. Stress	3	5		E	4/5/2021 8:29 AM	E	9/7/2021 5:59 PM
79. Fear of consequences/ complaint and legal action	2	2		E	4/5/2021 2:37 PM	E	9/7/2021 6:00 PM
80. Information (lack of it or quality of it)	8	8		E	3/5/2021 11:52 AM	E	9/7/2021 6:01 PM
81. Meta-ignorance	1	1		E	8/5/2021 5:08 PM	E	9/7/2021 3:14 PM
82. Context	1	8		E	23/5/2021 9:06 AM	E	9/7/2021 6:01 PM
83. Lack of clarity or support or help from other peers	1	1		E	26/5/2021 6:07 PM	E	9/7/2021 6:02 PM
84. Language and Understanding	1	1		E	2/5/2021 3:02 PM	E	9/7/2021 6:04 PM
85. Use analysing skills to reduce uncertainty	1	1		E	8/5/2021 4:16 PM	E	9/7/2021 3:15 PM
86. Complexity	5	8		E	20/5/2021 4:31 PM	E	9/7/2021 3:16 PM
87. Networks and Peers to reduce uncertainty	1	1		E	8/5/2021 4:32 PM	E	9/7/2021 6:06 PM
88. Conflict and disagreement with patient	1	1		E	25/5/2021 9:49 AM	E	9/7/2021 6:06 PM

Audit trail – Thematic synthesis on uncertainties in healthcare

Initial coding in N-vivo – 107 codes identified							
Name	**Files**	**References**	**Memo Link**	**Created By**	**Created On**	**Modified By**	**Modified On**
89. Convincing and communicating skills	1	1		E	20/5/2021 4:28 PM	E	9/7/2021 3:16 PM
90. Treatment choices led by fear, helplessness and frustration from uncertainty	1	1		E	20/5/2021 4:29 PM	E	9/7/2021 6:07 PM
91. People's belief and values	3	3		E	25/5/2021 12:28 PM	E	9/7/2021 6:08 PM
92. Probabilistic nature or ambiguity of the issu	1	1		E	25/5/2021 6:28 PM	E	9/7/2021 6:08 PM
93. Practical uncertainties	1	2		E	25/5/2021 11:05 AM	E	9/7/2021 6:09 PM
94. People's subjective interpretation of information	2	3		E	20/5/2021 4:29 PM	E	9/7/2021 6:10 PM
95. Patient's ambiguity	1	2		E	20/5/2021 7:35 PM	E	9/7/2021 6:11 PM
96. Cognitive experience of individuals	2	2		E	26/5/2021 6:12 PM	E	9/7/2021 6:11 PM
97. Multiple layered nature of uncertainty	2	4		E	8/5/2021 4:34 PM	E	9/7/2021 3:17 PM
98. Leadership of complex situation/team	4	4		E	8/5/2021 4:35 PM	E	9/7/2021 6:12 PM
99. Stochastic uncertainty or measurement error	3	3		E	8/5/2021 4:35 PM	E	9/7/2021 6:13 PM
100. Multiple 'correct' values	1	1		E	25/5/2021 8:53 AM	E	9/7/2021 6:14 PM
101. Scientific uncertainty	3	3		E	25/5/2021 4:03 PM	E	8/7/2021 4:01 PM
102. Parameter uncertainty	1	1		E	28/5/2021 3:26 PM	E	9/7/2021 7:07 PM
103. Model uncertainty	1	1		E	28/5/2021 3:18 PM	E	8/7/2021 3:18 PM
104. Linguistic imprecision	2	3		E	24/5/2021 11:17 AM	E	9/7/2021 7:07 PM
105.Social arrangement of power or institutional configuration	1	1		E	25/5/2021 9:28 AM	E	9/7/2021 7:08 PM
106. Conceptual uncertainty	1	2		E	25/5/2021 9:27 AM	E	9/7/2021 7:08 PM
107. Decision on behalf of child	1	1		E	25/5/2021 9:27 AM	E	9/7/2021 7:08 PM

Audit trail – Thematic synthesis on uncertainties in healthcare

Final 77 Codes from 20 articles classifying uncertainties using empirical research and case studies



Audit trail – Thematic synthesis on uncertainties in healthcare

45. Influence of friends and family
46. Patient's level of education
47. Patients' personal beliefs
48. Patients' personal fears
49. Risk taking behaviour
50. Trust issues
51. Uncertainty negatively perceived
52. Uncertainty positively perceived (coping with)
53. Proxy uncertainties (Carers)
54. Associated fears of parents
55. Personal interpretations (parents)

Stakeholder related

56. Lack of quantifiability or measurability
57. Model inadequacy
58. Procedural uncertainties
59. Guidelines based on evidence which cannot be generalised
60. Inability to apply guidelines to all situations

Related to procedures, guidelines

61. Uncertainties due to socio-economic inequalities and inequities
62. Uncertainties in implementing preventive measures
63. Uncertainty due to lack of standardised definitions
64. Uncertainties due to challenges in data collection
65. Uncertainties due to previous faulty reporting
66. Uncertainty due to unclear risk distribution

At the level of public health

67. Interaction of clinicians with other health care team leading to uncertainties
68. Interaction or conflict between patients' friends, relatives with clinician team leading to uncertainties
69. Physician- patient interaction leading to uncertainties
70. Improper team management leading to uncertainties
71. Trust issues in the team leading to uncertainties
72. Uncertainties due to avoiding referral
73. Uncertainties due to involvement of multiple people
74. Uncertainties due to referral
75. Uncertainties due to seeking help from others
76. Fear of stigma leading to uncertainties
77. Helplessness or feeling unsupported leading to uncertainties

Interaction between health care team, patients, carers, family, society

Audit trail – Thematic synthesis on uncertainties in healthcare

Descriptive themes:

1. Descriptive theme- Stochastic uncertainty
2. Descriptive theme- Diagnostic uncertainties
3. Descriptive theme- Prognostic uncertainties
4. **Descriptive theme- Communication of knowledge or information**
5. **Descriptive theme- Knowledge related to stakeholders**
6. **Descriptive theme- Quality of knowledge or information**
7. **Descriptive theme- Quantity of knowledge or information**
8. Descriptive theme- Ethical dilemmas
9. Descriptive theme- Proxy uncertainties
10. **Descriptive theme- lack of evidence in new diseases**
11. **Descriptive theme- Uncertainties due to stakeholders involved in key decision making**
12. **Descriptive theme- Uncertainties which are disease related (global emergency related**)
13. Descriptive theme- Contextual uncertainties
14. Descriptive theme- Investigative uncertainties
15. Descriptive theme- Lack of resources
16. Descriptive theme-Uncertainties related to doctors
17. Descriptive theme- Uncertainties related to patients
18. Descriptive theme- Uncertainty related to carers (Parents, children)
19. Descriptive theme- Methodological uncertainties
20. Descriptive theme-Uncertainty pertaining to guidelines
21. Descriptive theme- Uncertainties in establishing the standards
22. Descriptive theme- Uncertainties related to data
23. Descriptive theme- Interpersonal relationship leading to uncertainties
24. Descriptive theme- Lack of unified teamwork leading to uncertainties
25. Descriptive theme- Networking leading to uncertainties
26. Descriptive theme- Social relationships leading to uncertainties

Analytical themes:

1. Analytical theme- Epistemic uncertainties
2. Analytical theme- Knowledge exchange related uncertainties
3. Analytical theme- Global (Political) uncertainties
4. Analytical theme- Public health uncertainties
5. Analytical theme- Health care system uncertainties
6. Analytical theme- Clinical uncertainties
7. Analytical theme- Parameter uncertainties
8. Analytical theme- Personal uncertainties
9. Analytical theme- Relational uncertainties
10. Analytical theme- Ethical uncertainties
11. Analytical theme- Aleatoric uncertainties

Audit trail – Thematic synthesis on uncertainties in healthcare

Segregation of codes into descriptive themes and descriptive themes into analytical themes		
Final codes	**Descriptive themes**	**Analytical themes**
Unpredictability leading to uncertainties	Stochastic uncertainty	Aleatoric uncertainties
Statistical uncertainty		
Future risk leading to uncertainties		
Expensive investigations	Diagnostic uncertainties	Clinical uncertainties
Lack of robust clinical protocols		
Resemblance to other diseases		
Role of co-factors	Prognostic uncertainties	
Seriousness of condition		
Negotiation skills	**Communication of knowledge or information**	Knowledge exchange related uncertainties
Uncertainties due to cues picked during conversation		
Uncertainties due to language used during communication		
Uncertainty due to lack of patient-centred communication		
Clinicians' ignorance or lack of skill	**Knowledge related to stakeholders**	Epistemic uncertainties
Patient's level of education		
Ambiguity leading to uncertainties	**Quality of knowledge or information**	
Complexity leading to uncertainties		
Lack of credibility leading to uncertainties		
Growth of medical technology related knowledge leading to uncertainties	**Quantity of knowledge or information**	
Less knowledge or information leading to uncertainties		
More knowledge or information leading to uncertainties		
Prioritising patients due to lack of resources	Ethical dilemmas	Ethical Uncertainties
Time pressure dictating patient treatment		
To inform or not to inform patients about lack of evidence		
Clinicians making decisions on behalf of the patient or family members	Proxy uncertainties	

Audit trail – Thematic synthesis on uncertainties in healthcare

Segregation of codes into descriptive themes and descriptive themes into analytical themes		
Final codes	**Descriptive themes**	**Analytical themes**
Suddenness of the situation	**lack of evidence in new diseases**	Global uncertainties
Uncertainty due to lack of conclusive results from emergency research		
Uncertainties created by media	**Uncertainties due to stakeholders involved in key decision making**	
Uncertainties due to social arrangement of power		
Uncertainties due to subjective decision making		
Inability to establish a cause-and-effect relationship	**Uncertainties which are disease related (global emergency related)**	
Uncertainties due to lack of pattern in disease presentation		
Timing of the week or day	Contextual uncertainties	Health care system uncertainties
Errors in reporting	Investigative uncertainties	
Over testing in response to or leading to uncertainties		
Lack of check mechanism	Lack of resources	
Lack of facilities		
Level of experience	Uncertainties related to doctors	Personal uncertainties
Clinicians' personal fears		
Previous experiences (clinician)		
Proxy uncertainties (clinician)		
Risk taking behaviour (Clinician)		
Uncertainty intolerance (Clinician)		
Unclear patient characteristics	Uncertainties related to patients	
Influence of friends and family		
Patient's level of education		
Patients' personal beliefs		

Audit trail – Thematic synthesis on uncertainties in healthcare

Segregation of codes into descriptive themes and descriptive themes into analytical themes		
Final codes	**Descriptive themes**	**Analytical themes**
Patients' personal fears		
Risk taking behaviour		
Trust issues		
Uncertainty negatively perceived		
Uncertainty positively perceived (coping with)		
Proxy uncertainties (Carers)	Uncertainties related to carers	
Associated fears of parents		
Personal interpretations (parents)		
Lack of quantifiability or measurability	Methodological uncertainties	Parameter uncertainties
Model inadequacy		
Procedural uncertainties		
Guidelines cannot be generalised	Uncertainty pertaining to guidelines	
Inability to apply guidelines to all situations		
Uncertainties due to socio-economic inequalities and inequities	Uncertainties in establishing and implementing the national standards	Public health uncertainties
Uncertainties in implementing preventive measures		
Uncertainty due to lack of standardised definitions		
Uncertainties due to challenges in data collection	Uncertainties related to epidemiological data	
Uncertainties due to lack of previous records		
Uncertainty due to unclear risk distribution		
Interaction of clinicians with other health care team leading to uncertainties	Interpersonal relationship leading to uncertainties	Relational uncertainties
Interaction or conflict between patients' friends, relatives with clinician team leading to uncertainties		
Physician- patient interaction leading to uncertainties		
Improper team management leading to uncertainties	Lack of unified teamwork	

Audit trail – Thematic synthesis on uncertainties in healthcare

Segregation of codes into descriptive themes and descriptive themes into analytical themes		
Final codes	**Descriptive themes**	**Analytical themes**
Trust issues in the team leading to uncertainties	leading to uncertainties	
Uncertainties due to avoiding referral	Networking leading to uncertainties	
Uncertainties due to involvement of multiple people		
Uncertainties due to referral		
Uncertainties due to seeking help from others		
Fear of stigma leading to uncertainties	Social relationships leading to uncertainties	
Helplessness or feeling unsupported leading to uncertainties		